

Generation Interconnection Request System Impact Study Report

For Queue Project Q18-1119

Southtown 345 kV
July 22, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1119. The Interconnection Customer proposes a 259.8 MW wind facility interconnecting at the Southtown 345 kV substation in the Maryland area. The request was received on November 30, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 350 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$12,644,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: PIT 6 - LAS LOMAS 138.0 kV
Contingency: VIENNA OPERATIONS SUBSTATION - EASTON 2 345.0 kV
Loading: 131.2% (Rating: 234 MVA)
DFAX: 10.1%
Pre-project loading: 119.9%
2. Facility: PRICE - WHITE BLUFF 138.0 kV
Contingency: CADY - EASTON 2 345.0 kV
Loading: 130.3% (Rating: 409 MVA)
DFAX: 12.7%
Pre-project loading: 122.2%
3. Facility: GALVANIZING - SANTEE LAH 138.0 kV
Contingency: SOUTHTON - VIENNA OPERATIONS 345.0 kV
Loading: 129.2% (Rating: 722 MVA)
DFAX: 22.3%
Pre-project loading: 121.2%
4. Facility: SANTEE LAH - SUNSET 138.0 kV
Contingency: VIENNA OPERATIONS SUBSTATION - SOUTHTOWN 345.0 kV
Loading: 128.8% (Rating: 730 MVA)
DFAX: 22.3%
Pre-project loading: 120.9%
5. Facility: HOLSTON DAM - NAVY MOLE 138.0 kV Line
Contingency: CADY SUBSTATION - EASTON 2 345.0 kV
Loading: 128.2% (Rating: 949 MVA)
DFAX: 25.3%
Pre-project loading: 121.2%

6. Facility: GRASONVILLE - CARANCAHUA 138.0 kV
Contingency: CADY - EASTON 2 345.0 kV
Loading: 128.0% (Rating: 227 MVA)
DFAX: 6.1%
Pre-project loading: 121.0%
7. Facility: ROBINSONVILLE - QUINN 138.0 kV
Contingency: VIENNA OPERATIONS - EASTON 2 345.0 kV L
Loading: 127.5% (Rating: 239 MVA)
DFAX: 8.2%
Pre-project loading: 118.6%
8. Facility: DONNER - ORANGE BULK 138.0 kV
Contingency: CADY - DODD RD 345.0 kV
Loading: 126.1% (Rating: 200 MVA)
DFAX: 7.5%
Pre-project loading: 116.5%
9. Facility: BOND-ELK GROVE FLORIN 345 kV/138 kV Transformer Bank
Contingency: DODD ROAD - CADY 345.0 kV
Loading: 126.1% (Rating: 922 MVA)
DFAX: 34.7%
Pre-project loading: 116.3%
10. Facility: ORANGE BULK - WHITE BLUFF 138.0 kV
Contingency: CADY - EASTON 2 345.0 kV L
Loading: 123.9% (Rating: 293 MVA)
DFAX: 10.2%
Pre-project loading: 114.9%
11. Facility: SOONER - KIWI 138.0 kV line
Contingency: SUNSET - SADDLEBROOKE RANCH 138.0 kV
Loading: 123.4% (Rating: 252 MVA)
DFAX: 9.1%
Pre-project loading: 114.0%
12. Facility: WHITMORE - BOND-ELK GROVE FLORIN 138.0 kV
Contingency: CADY SUBSTATION - DODD RD SUBSTATION 345.0 kV
Loading: 122.2% (Rating: 274 MVA)
DFAX: 9.4%
Pre-project loading: 113.3%
13. Facility: WILLOW PASS - BANTA CARBONA 138.0 kV
Contingency: DODD RD - CADY SUBSTATION 345.0 kV
Loading: 119.8% (Rating: 354 MVA)
DFAX: 12.1%
Pre-project loading: 110.9%
14. Facility: VERNON - LIMON 138.0 kV
Contingency: CADY - EASTON 2 345.0 kV
Loading: 118.8% (Rating: 512 MVA)
DFAX: 13.3%
Pre-project loading: 112.1%
15. Facility: ESS LEWES - REHOBOTH 138.0 kV
Contingency: VIENNA OPERATIONS - EASTON 2 345.0 kV
Loading: 118.8% (Rating: 646 MVA)
DFAX: 25.0%
Pre-project loading: 108.7%
16. Facility: LAMMERS SUBSTATION - SADDLEBROOKE RANCH 138.0 kV L
Contingency: VIENNA OPERATIONS - EASTON 2 345.0 kV L

- Loading: 118.2% (Rating: 1,304 MVA)
DFAX: 47.7%
Pre-project loading: 108.6%
17. Facility: BOND-ELK GROVE FLORIN - FAIRMONT 138.0 kV Cir. 1
Contingency: CADY - DODD RD 345.0 kV line
Loading: 118.0% (Rating: 477 MVA)
DFAX: 15.0%
Pre-project loading: 109.9%
18. Facility: FAIRMONT SUBSTATION - BOND-ELK GROVE FLORIN 138.0 kV Line 2
Contingency: CADY - DODD RD 345.0 kV
Loading: 118.0% (Rating: 477 MVA)
DFAX: 15.0%
Pre-project loading: 109.9%
19. Facility: BUCKSKIN - VERNON 138.0 kV
Contingency: CADY - EASTON 2 345.0 kV
Loading: 118.0% (Rating: 525 MVA)
DFAX: 13.3%
Pre-project loading: 111.4%
20. Facility: N LEBANON - FAIRMONT 138.0 kV Ckt 1
Contingency: CADY - DODD ROAD 345.0 kV
Loading: 117.6% (Rating: 484 MVA)
DFAX: 15.0%
Pre-project loading: 109.5%
21. Facility: NORTH LEBANON - FAIRMONT 138.0 kV
Contingency: CADY - DODD RD 345.0 kV
Loading: 117.6% (Rating: 484 MVA)
DFAX: 15.0%
Pre-project loading: 109.5%
22. Facility: BUCKSKIN - ALUM RIDGE 138.0 kV
Contingency: EASTON 2 - CADY 345.0 kV
Loading: 117.3% (Rating: 538 MVA)
DFAX: 13.3%
Pre-project loading: 110.9%
23. Facility: DOBBINS - EASTON 138.0 kV line
Contingency: CADY - EASTON 2 SUBSTATION 345.0 kV
Loading: 116.7% (Rating: 474 MVA)
DFAX: 18.9%
Pre-project loading: 106.3%
24. Facility: ALUM RIDGE - HILLSBORO 138.0 kV
Contingency: EASTON 2 - CADY 345.0 kV L
Loading: 116.1% (Rating: 559 MVA)
DFAX: 13.3%
Pre-project loading: 110.0%
25. Facility: RIPPLE 345/138kV XFMR
Contingency: VIENNA OPERATIONS - SOUTHTOWN 345.0 kV
Loading: 116.0% (Rating: 907 MVA)
DFAX: 34.8%
Pre-project loading: 106.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
PIT 6 - LAS LOMAS 138.0 kV Line	Rebuild line with higher-capacity conductor	\$11,630,000	6% (\$657,000)
PRICE SUBSTATION - WHITE BLUFF 138.0 kV	Rebuild line with higher-capacity conductor	\$26,145,000	6% (\$1,441,000)
GALVANIZING - SANTEEHLAH 138.0 kV	Rebuild line with higher-capacity conductor	\$5,869,000	6% (\$340,000)
SUNSET - SANTEEHLAH 138.0 kV	Rebuild line with higher-capacity conductor	\$3,183,000	6% (\$185,000)
HOLSTON DAM - NVY MOLE 138.0 kV	Rebuild line with higher-capacity conductor	\$4,532,000	6% (\$252,000)
GRASONVILLE - CARANCAHUA 138.0 kV	Rebuild line with higher-capacity conductor	\$5,823,000	6% (\$327,000)
ROBINSONVILLE - QUINN 138.0 kV	Rebuild line with higher-capacity conductor	\$5,682,000	6% (\$359,000)
DONNER - ORANGE BULK 138.0 kV Line	Rebuild line with higher-capacity conductor	\$40,087,000	5% (\$2,068,000)
BOND-ELK GROVE FLORIN 345 kV/138 kV Transformer	Replace transformer with higher-rated unit	\$15,140,000	5% (\$769,000)
ORANGE BULK - WHITE BLUFF SUBSTATION 138.0 kV	Rebuild line with higher-capacity conductor	\$12,266,000	5% (\$643,000)
SOONER - KIWI 138.0 kV line	Rebuild line with higher-capacity conductor	\$3,836,000	6% (\$211,000)
WHITMORE - BOND-ELK GROVE FLORIN 138.0 kV	Rebuild line with higher-capacity conductor	\$3,719,000	5% (\$189,000)
WILLOW PASS - BANTA CARBONA 138.0 kV	Rebuild line with higher-capacity conductor	\$24,069,000	5% (\$1,247,000)
VERNON - LIMON 138.0 kV L	Rebuild line with higher-capacity conductor	\$4,776,000	5% (\$254,000)
ESS LEWES - REHOBOTH 138.0 kV	Rebuild line with higher-capacity conductor	\$6,696,000	6% (\$416,000)
SADDLEBROOKE RANCH - LAMMERS 138.0 kV	Rebuild line with higher-capacity conductor	\$11,261,000	6% (\$678,000)
BOND-ELK GROVE FLORIN - FAIRMONT 138.0 kV Cir. 1	Rebuild line with higher-capacity conductor	\$2,393,000	5% (\$119,000)
FAIRMONT SUBSTATION - BOND-ELK GROVE FLORIN 138.0 kV Line 2	Rebuild line with higher-capacity conductor	\$2,268,000	5% (\$113,000)
BUCKSKIN - VERNON 138.0 kV	Rebuild line with higher-capacity conductor	\$4,583,000	5% (\$244,000)
N LEBANON - FAIRMONT 138.0 kV Ckt 1	Rebuild line with higher-capacity conductor	\$3,566,000	5% (\$177,000)
NORTH LEBANON - FAIRMONT 138.0 kV Ckt 2	Rebuild line with higher-capacity conductor	\$3,753,000	5% (\$186,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
ALUM RIDGE - BUCKSKIN 138.0 kV	Rebuild line with higher-capacity conductor	\$7,860,000	5% (\$418,000)
DOBBINS - EASTON 138.0 kV line	Rebuild line with higher-capacity conductor	\$2,846,000	6% (\$158,000)
ALUM RIDGE - HILLSBORO SUBSTATION 138.0 kV	Rebuild line with higher-capacity conductor	\$6,998,000	5% (\$372,000)
RIPPLE 345-138 kV Autotransformer	Replace transformer with higher-rated unit	\$12,667,000	6% (\$821,000)
	Total allocated to project		\$12,644,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.